AP PRECALCULUS

Unit 1 • Worksheet 1.13-1.14A • Answer key

Worksheet 1.13-1.14A • Function Model Construction

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Data with equally spaced inputs have first differences $4.9,\ 5.1,\ 5.0,\ 4.9$. Which model fits best?

(A) linear
(B) quadratic
(C) cubic
(D) rational

Solution

Answer: (A).

Question 2

NO CALCULATOR

A company's average cost per item is $A(n) = \frac{500+4n}{n}$. What does $\lim_{n\to\infty} A(n)$ tell you?

(A) The fixed cost is \$500.
(B) The average cost becomes 0.
(C) The average cost per item approaches \$4.
(D) The cost per item grows without bound.

Solution

Answer: (C).

Question 3

NO CALCULATOR

A rectangle has perimeter 40 cm and width w. Which gives its area?

(A) $w(40 - w)$
(B) $w(20 - w)$
(C) $2w + 40$
(D) $w^2 - 20$

Solution

Answer: (B).

Question 4

CALCULATOR

The height of a model rocket is recorded: $(0, 1.2)$, $(1, 6.1)$, $(2, 9.0)$, $(3, 9.9)$, $(4, 8.6)$ (t in s, h in m). Using quadratic regression, what is the predicted height at $t = 5$?

(A) 4.2 m
(B) 5.3 m
(C) 6.4 m
(D) 7.3 m

Solution
Answer: (B). $h(t) \approx -1.029t^2 + 5.974t + 1.183$; $h(5) \approx 5.34$.

Question 5 (Free response)

CALCULATOR

A cylindrical can holds 500 cm^3. Let r be the radius and h the height (cm).

(a) Express h in terms of r, and show that the surface area is $S(r) = 2\pi r^2 + \frac{1000}{r}$.
(b) Find the radius that minimizes S, and the minimum surface area.
(c) Describe S as $r \to 0^+$ and $r \to \infty$, and explain in context.

Solution
(a) $h = \frac{500}{\pi r^2}$; $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{1000}{r}$.
(b) $r \approx 4.301$ cm; $S \approx 348.734$ cm^2.
(c) $S \to \infty$ in both cases: a very thin can must be very tall; a very wide can has a huge top and bottom.

Question 6 (Free response)

NO CALCULATOR

A phone plan costs \$20 per month including 100 texts, plus \$0.10 per extra text.

(a) Write a piecewise model $C(n)$ and find $C(250)$.
(b) For $n > 100$, find $\lim_{n\to\infty} \frac{C(n)}{n}$ and interpret it.

Solution
(a) $C(n) = 20$ for $0 \le n \le 100$; $20 + 0.1(n - 100)$ for $n > 100$. $C(250) = 35$.
(b) $\frac{10+0.1n}{n} \to 0.10$: the average cost per text approaches \$0.10.